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To cite this article: Yaowarat Sirisathitkul, Natthinee Dinmeung, Wannasan Noonsuk & Chitnarong Sirisathitkul (2025) Accuracy and precision of smartphone colorimetry: a comparative analysis in RGB, HSV, and CIELAB color spaces for archaeological research, STAR: Science & Technology of Archaeological Research, 11:1, e2444168, DOI: [10.1080/20548923.2024.2444168](https://doi.org/10.1080/20548923.2024.2444168)

To link to this article: <https://doi.org/10.1080/20548923.2024.2444168>



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Published online: 20 Jan 2025.



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Accuracy and precision of smartphone colorimetry: a comparative analysis in RGB, HSV, and CIELAB color spaces for archaeological research

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ABSTRACT

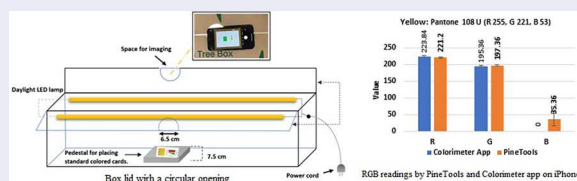
This study evaluates the accuracy and precision of color measurements obtained from digital images captured using smartphone cameras, validated against standard color cards under controlled lighting conditions. RGB (Red, Green, Blue) and HSV (Hue, Saturation, Value) parameters were extracted using the Colorimeter application (app) on an iPhone and the PineTools program on a computer. Both methods demonstrated high precision, as indicated by small standard deviations in repeated measurements. Significant deviations from Pantone reference values were also observed in the overestimation of S and underestimation of V values. Notably, the Color Grab app on an Android phone exhibited superior accuracy in extracting H and V values compared to the iPhone. Additionally, the CIELAB parameters measured by the Android app were comparable to those obtained from a portable colorimeter. These findings underscore the utility of smartphone colorimetry for precise on-site color classification, with significant potential applications in cultural heritage and archaeology.

ARTICLE HISTORY

Received 17 September 2024
Accepted 15 December 2024

KEYWORDS

Digital image; colorimetry; smartphone; color space; archaeology



1. Introduction

Colorimeters, traditionally employed in quality control within the textile and food industries, have expanded their applications across various fields, including chemical and biological analysis (Ma et al. 2018; Wu et al. 2023), as well as cultural heritage preservation and archaeology (Capobianco et al. 2021; Fahim, Ghasemi, and Hosseini-Zori 2023; Milotta et al. 2018; Sanmartín et al. 2024). These instruments quantify color by analyzing the absorption and reflection of light across wavelengths in the visible spectrum. A wide range of colors can be represented in the RGB color space, where red, green, and blue light are combined in various intensities. In electronic systems, color is expressed as RGB values, each ranging from zero to a maximum value (e.g. 255 for 8-bit per color channel). To address specific application needs, alternative color spaces such as CIELAB and HSV (Hue, Saturation, Value of brightness) are often employed (Fan et al. 2021; Nelis et al.

2019). The HSV color space closely resembles human perception of color and provides an intuitive cone-shaped representation of color variations (Fan et al. 2021; Rani et al. 2021). Hue (H), measured in degrees (0–360), represents the actual color (e.g. red, blue, or yellow). Saturation (S), ranging from 0% (gray or white) to 100% (vibrant color), reflects color purity or vibrancy, with higher saturation indicating more intense colors. Value of brightness (V), also ranging from 0% (black) to 100% (brightest color), represents the color's brightness. The CIELAB color space, commonly used in food characterizations, defines color based on three parameters: L* (lightness, ranging from 0 to 100), a* (greenness to redness, ranging from negative to positive), and b* (blueness to yellowness, ranging from negative to positive) (Fan et al. 2021; Gueli et al. 2019).

Digital image colorimetry quantifies colors without requiring specialized equipment or lengthy

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procedures. RGB, HSV, and CIELAB color parameters can be extracted from images captured by digital cameras, scanners, or other color-recording devices with appropriate software on a personal computer (Apyari et al. 2017). Furthermore, the widespread use of smartphones presents additional opportunities for digital image colorimetry (Grossi 2019). With built-in cameras and colorimetric applications (apps) (e.g. Colorimeter, Catch Color, Color Grab, and Color Detector), color measurement is now more accessible and affordable for a wide range of users. Due to the importance of color in analytical chemistry, digital image colorimetry based on smartphones has been widely implemented in various chemical analyses, as reviewed by Fan et al. (2021). Additionally, smartphone colorimetry opens new possibilities in other fields where on-site measurements and point-of-care testing are desirable (Alawsi et al. 2021; Dinmeung, Sirisathitkul, and Sirisathitkul 2023; Qain et al. 2022). Other applications include evaluating the skin colors of animals (Cugmas and Štruc 2020) and humans (Hong and Kim 2019; Mulcare and Coward 2019) and assessing soil colors as an alternative to traditional Munsell charts (Fan et al. 2017; Gómez-Robledo et al. 2013). In archaeology, smartphone colorimetry has been applied to color characterizations of potsherds from maritime Southeast Asia (Sirisathitkul et al. 2020) and lithic assemblages from the Spanish La Calvera rock shelter (Ramacciotti et al. 2023).

In contrast to spectroscopic techniques that measure photon absorption at specific wavelengths, digital image colorimetry operates within the visible light range but does not comprehensively measure absorption across the entire range of 380–720 nm due to the constraints imposed by the spectral sensitivity of the camera sensor (Trombini et al. 2021). Charge-Coupled Device (CCD) and Complementary Metal Oxide Semiconductor (CMOS) image sensors exhibit non-uniform spectral sensitivity, and color readings may vary with exposure levels, white balance, and other camera settings. The method provides an approximation based on how colors are represented in the RGB, HSV, or CIELAB color spaces. In smartphone colorimetry, the types of smartphone cameras and colorimetric apps used significantly influence the results (Ciaccheri et al. 2023; Fan et al. 2021; Rani et al. 2021). Furthermore, environmental factors such as lighting conditions and other measurement conditions can impact the accuracy of measurements (Gueli et al. 2019; Plutino and Simone 2021). Calibration and ambient subtraction are thus recommended to ensure reliable results (Nixon, Outlaw, and Leung 2020).

To evaluate the accuracy and precision of color measurement, standard color cards can be employed as a color calibration target. Previous studies by You et al. (2020) demonstrated the use of color cards in

standardizing smartphone colorimetry for chicken meat analysis. Similarly, Yoo et al. (2021) utilized the PicMan software to analyze RGB, HSV, CIELAB, and Munsell colors of Gouache paint color cards before proceeding to the cultural heritage examination. This study explores comparative smartphone colorimetry using both iPhone and Android phone. Reference color cards with standardized Pantone values were utilized under controlled lighting conditions to evaluate the accuracy and precision of the measurements. The discussion is organized into four subsections, focusing on the RGB, HSV, and CIELAB color spaces, followed by examining the implications for archaeological research.

2. Materials and methods

This study employed color cards from the Color Mixer Accessory Kit OS-8495 (Pasco, USA), which included the following colors: Black, White, Gray, Red, Green, Blue, Yellow, Orange, Pink, and Purple (Figure 1). The Pantone color values for these cards were obtained from the iColorpalette website (<https://icolorpalette.com/>) and used as reference values. To ensure consistent lighting conditions, each color card was placed inside a closed cardboard box ($30 \times 100 \times 30 \text{ cm}^3$) equipped with a daylight Light Emitting Diode (LED) lamp (NPV EVO JL6A-9W, 6500 K) as the light source. This setup minimized the impact of external factors such as ambient light and camera-to-object distance. Colorimetric measurements were performed using both iPhone and Android phone.

2.1. RGB and HSV measurements using the PineTools program and the Colorimeter app on an iPhone

An iPhone XR (12 MP camera, f/1.8 aperture) was positioned over an opening at the top of the box, maintaining a fixed distance of 22.5 cm between the camera lens and the color card, as illustrated in Figure 2. Under the controlled LED daylight conditions, images of the color cards were captured. The Colorimeter app (developed by Serhil Smyk) was used to extract RGB and HSV values directly on the smartphone. In addition, the images (saved as .png files at a resolution of 3024×3032 pixels) were transferred to a personal computer for further analysis using the PineTools program. Available at <https://pinetools.com>, PineTools is an online toolset that facilitates the extraction of color parameters. For each card, five measurements were repeated on five different regions: the center, top left, top right, bottom left, and bottom right. The average values and standard deviations were calculated to evaluate consistency. To evaluate accuracy, the measurements obtained with iPhone's Colorimeter app and

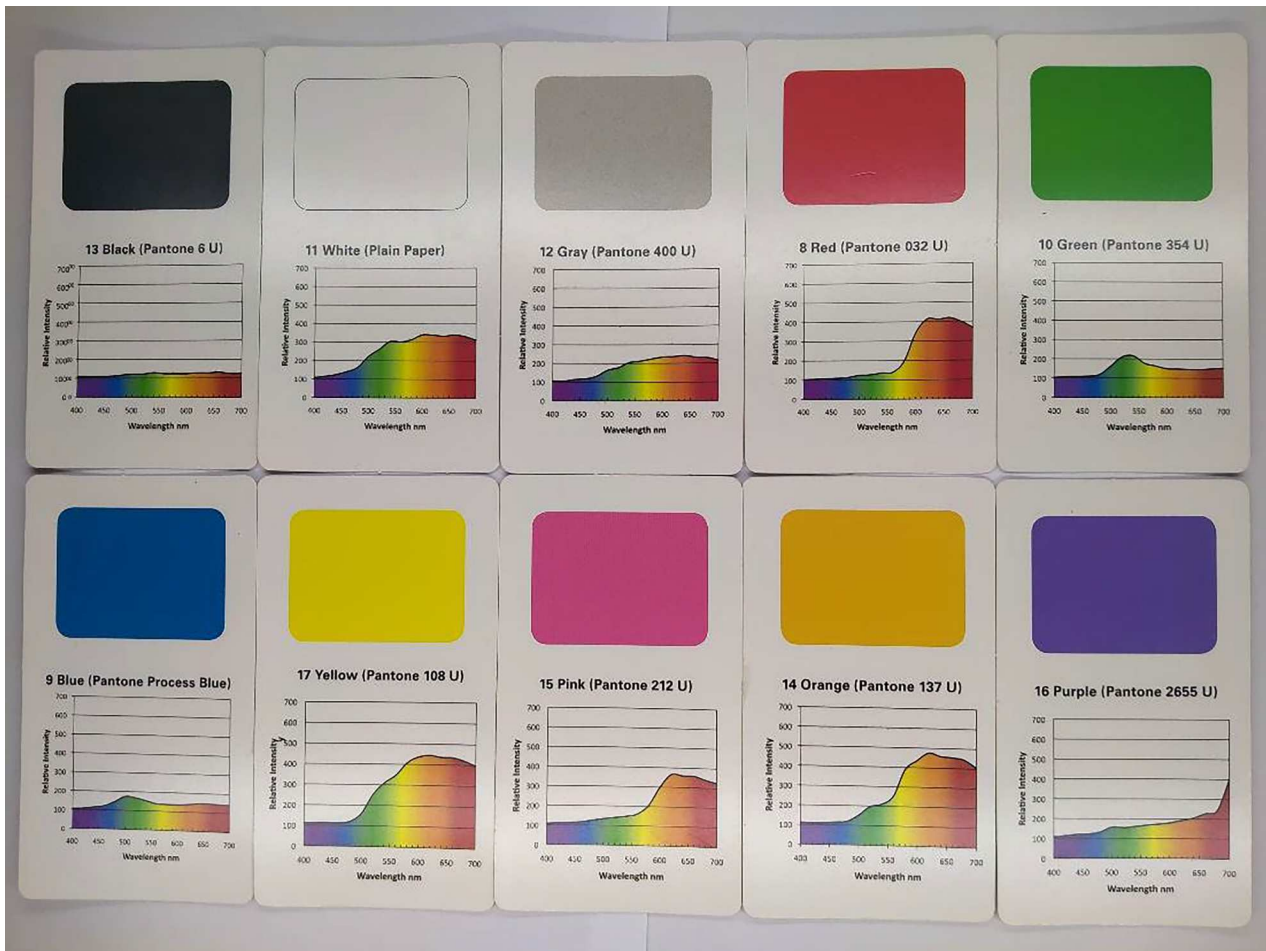


Figure 1. Ten cards from the Color Mixer Accessory Kit (Pasco OS-8495) used in this study.

the PineTools program on a computer were compared with the Pantone reference values.

using the following equation.

$$\Delta E_{ab}^* = \sqrt{(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2} \quad (1)$$

2.2. $L^*a^*b^*$ and HSV measurements using a portable colorimeter and the Color Grab app on an Android phone

Colorimetric measurements were also conducted using an Android phone (Oppo A12; 13 MP camera, f/2.2 aperture) in combination with the Color Grab app (Loomatix). This app enabled rapid HSV and CIELAB color analysis of the captured images by Android phones. A portable colorimeter (NH310, 3nh, China) was also employed to ensure accurate benchmarking. The device measured $L^*a^*b^*$ values by capturing reflected colors from a cross-sectional area of 8 mm in diameter under LED blue light excitation. The colorimeter utilized an Illuminating/Viewing Geometry of 8/d for precise reflectance measurements. The dual analysis with the Pantone reference values from the iColor-palette website allowed a comparative assessment for validating the smartphone-based methods. To determine the difference between the reference and the measured color, the ΔE_{ab}^* values were then calculated

3. Results and discussion

3.1. RGB readings by the PineTools program and the Colorimeter app on an iPhone

Figures 3 and 4 compare the average RGB readings by the Colorimeter app on the iPhone and the PineTools program on a computer. Except for the White card, the reference values for the nine cards are also provided. According to Figure 3(a and b), the RGB values from the Black and White cards obtained from both methods differ within the measurement uncertainty. For the Black card, minimal R, G, and B values were observed, consistent with the absorption across the visible light spectrum. Conversely, the White card shows substantial R, G, and B values from the reflection. The Colorimeter app readings ($R = 191.88$, $G = 198.88$, $B = 177.16$) are slightly lower than those from PineTools by only 1.25%, 0.12%, and 1.42%, respectively. For the Gray card (Figure 3(c)), reduced RGB values were observed in both methods compared to the White card, as

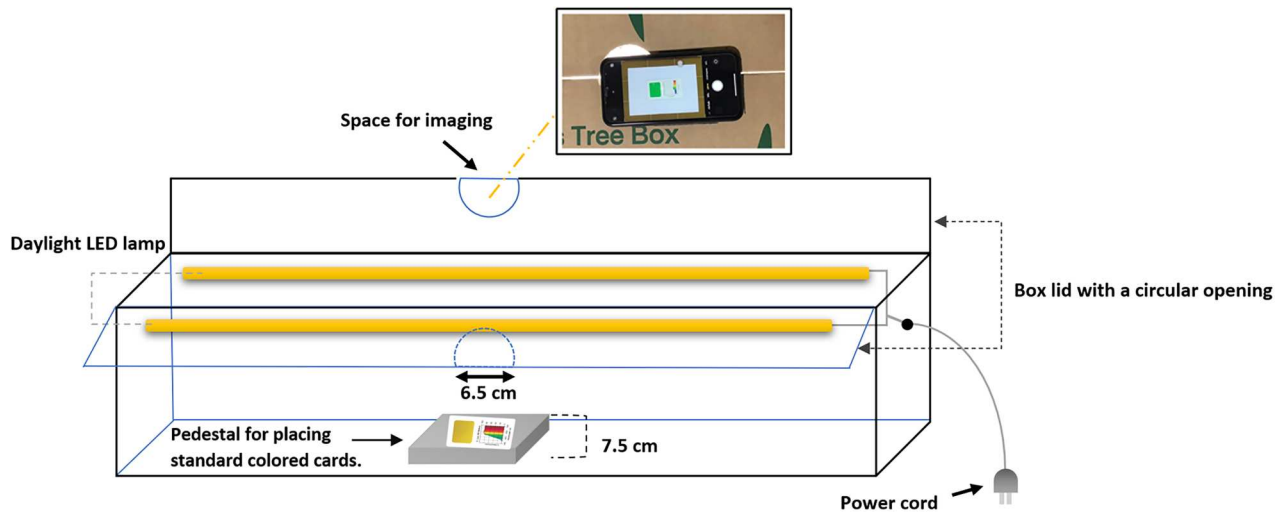


Figure 2. Experimental set-up for smartphone colorimetry. Settings on the smartphone in the inset are Photo mode (STD), White balance (Auto), Color effect (None), and Focus (Auto).

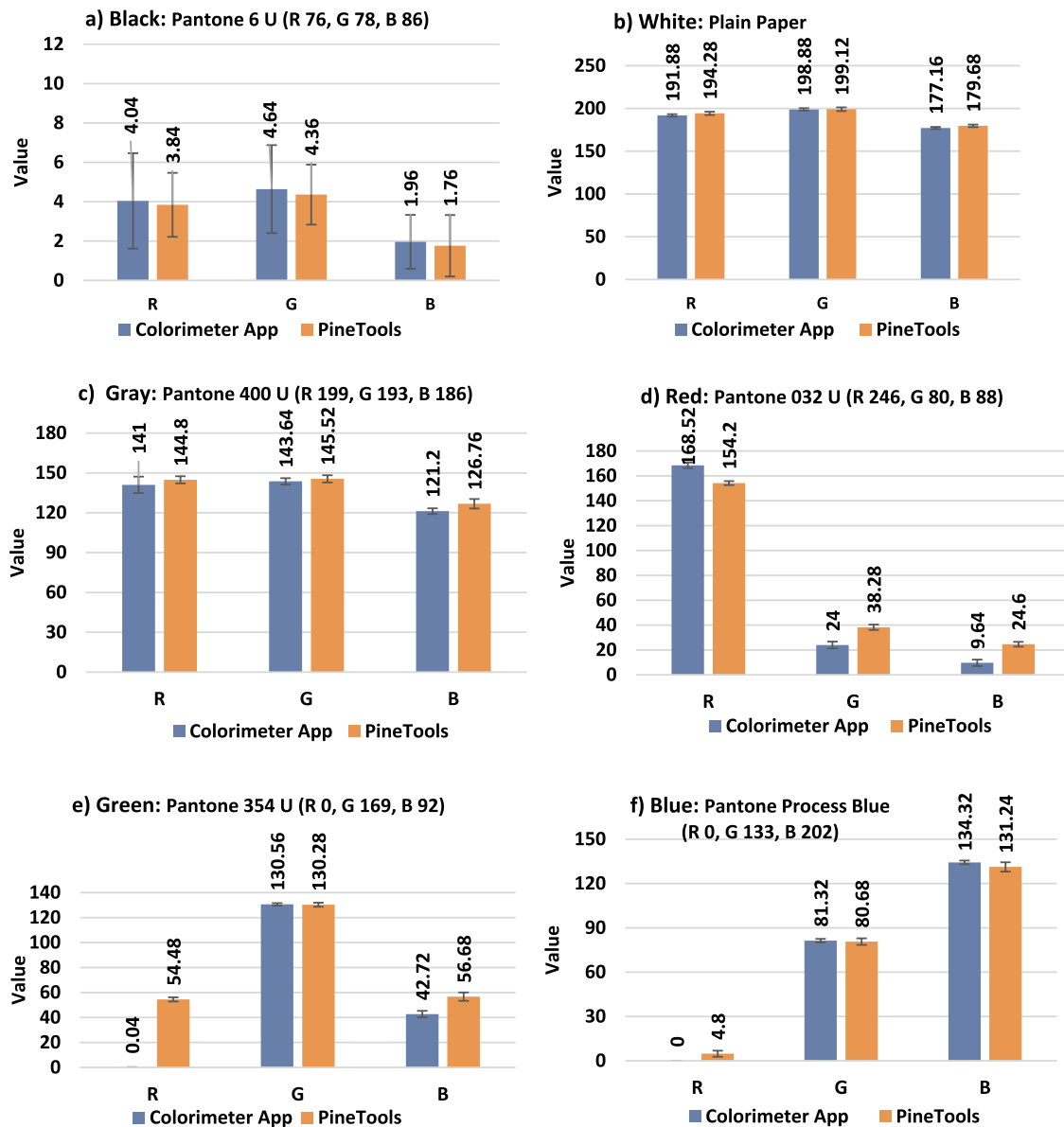


Figure 3. RGB colors of (a) Black; (b) White; (c) Gray; (d) Red; (e) Green; (f) Blue cards read by the Colorimeter app on a smartphone and the PineTools program on a computer.

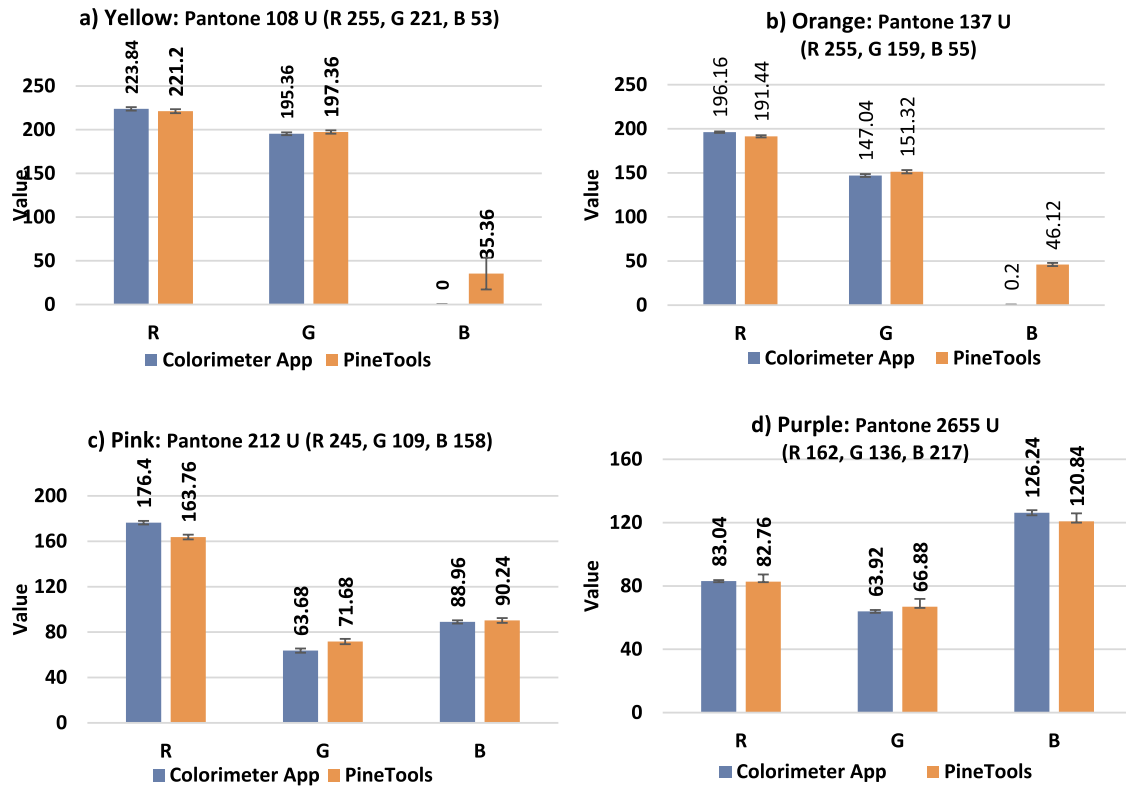


Figure 4. RGB colors of (a) Yellow; (b) Orange; (c) Pink; (d) Purple cards read by the Colorimeter app on a smartphone and the PineTools program on a computer.

expected. Like the Black and White cards, the R and G values are comparable, while the B value is consistently lower. This trend aligns with the reference values (R = 199, G = 193, B = 186).

For the Red, Green, and Blue cards (Figure 3(d–f)), the results reflect the dominance of the primary color in each card. The G values from the Green card and B values from the Blue card range from 130.28 to 134.32 for both methods. The R values measured from the Red card are higher (168.52 from the Colorimeter app and 154.2 from PineTools). However, PineTools tends to extract higher values for secondary color components, such as the B values for the Red (24.60 vs. 9.64) and Green (56.68 vs. 42.72) cards. Notably, for the Green and Blue cards, the R values measured by the Colorimeter app are near zero, consistent with the reference values. However, the PineTools program assigns a significant value to this component. These results from the iPhone suggest that RGB measurements using the Colorimeter app offer superior accuracy compared to PineTools.

The Yellow, Orange, Pink, and Purple cards demonstrate colors resulting from the primary color mixing of R, G, and B components. The differences between the two methods are not pronounced. The Yellow and Orange cards (Figure 4(a and b)) similarly show dominant R values and slightly lower G values from both methods, consistent with the reference values. The R component is still a major contributor to the Pink card (Figure 4(c)). The minor G component in the

Pink card shows a difference of 12.56% between methods, while this difference reduces to 4.63% for the Purple card (Figure 4(d)). B values become the major color component in the Purple card, with averages of 126.24 (Colorimeter app) and 120.84 (PineTools). However, these comparable values substantially differ from the reference. Despite the small standard deviations in all color cards in Figures 3 and 4, indicating high precision, the discrepancies from the Pantone reference values highlight the inaccuracy of RGB measurements using the Colorimeter app and PineTools.

3.2. HSV readings by the Colorimeter app on an iPhone and the Color Grab app on an Android phone

The minimal standard deviations observed in the repeated measurements of S (Table 1), V (Table 2), and H (not shown here) confirm the high precision of colorimetry conducted using both iPhone and Android phone. However, similar to the results in RGB measurements presented in Section 3.1, the deviations in HSV readings from the reference values indicate a lack of accuracy in smartphone colorimetry despite their excellent precision. Nevertheless, the H values measured across all methods are consistent and comparable to the Pantone references. Figure 5(a) presents scatter plots comparing measurements from the Colorimeter app against PineTools. The

Table 1. S values measured by the PineTools program on a computer, the Colorimeter app on the iPhone and the Color Grab app on the Android phone.

Card color	S (%)			
	Reference value	Colorimeter app on iPhone	PineTools	Color Grab app on Android phone
Black: Pantone 6 U	12	63.16 ± 20.63	65.08 ± 24.42	100 ± 0.00
White: Plain Paper	–	79.28 ± 0.37	74.82 ± 0.37	3.00 ± 0.00
Gray: Pantone 400 U	7	15.80 ± 0.50	13.12 ± 1.81	3.53 ± 0.52
Red: Pantone 032 U	67	94.24 ± 1.59	84.12 ± 1.24	85.53 ± 0.74
Green: Pantone 354 U	100	99.96 ± 0	58.64 ± 0.86	84.40 ± 2.92
Blue: Pantone Process Blue	100	100 ± 0	96.32 ± 1.49	100 ± 0.00
Yellow: Pantone 108 U	79	100 ± 0	83.92 ± 8.25	96.73 ± 0.46
Pink: Pantone 212 U	78	99.80 ± 0.41	75.88 ± 0.73	47.60 ± 1.30
Orange: Pantone 137 U	56	87 ± 0.96	56.12 ± 1.17	88.47 ± 1.36
Purple: Pantone 2655 U	37	63.88 ± 0.88	44.72 ± 1.90	31.47 ± 2.00

Note: The values from the iColorpalette website are also listed as reference.

Table 2. V values measured by the PineTools program on a computer, the Colorimeter app on the iPhone and the Color Grab app on the Android phone.

Card color	V (%)			
	Reference value	Colorimeter app on iPhone	PineTools	Color grab app on Android phone
Black: Pantone 6 U	34	1.76 ± 0.88	1.72 ± 0.68	1.20 ± 0.56
White: Plain Paper	–	11.32 ± 2.25	9.88 ± 0.33	91.67 ± 1.05
Gray: Pantone 400 U	78	56.28 ± 1.06	57.12 ± 1.09	74.80 ± 0.86
Red: Pantone 032 U	96	66.12 ± 0.88	60.48 ± 0.65	81.20 ± 1.21
Green: Pantone 354 U	66	51.2 ± 0.41	51.16 ± 0.62	63.60 ± 1.40
Blue: Pantone Process Blue	79	52.84 ± 0.47	51.64 ± 0.99	71.33 ± 0.98
Yellow: Pantone 108 U	100	87.8 ± 0.87	86.60 ± 1.00	91.60 ± 1.24
Pink: Pantone 212 U	100	76.80 ± 0.50	75.24 ± 0.44	87.13 ± 1.06
Orange: Pantone 137 U	96	69.16 ± 0.69	64.20 ± 0.71	90.60 ± 1.64
Purple: Pantone 2655 U	85	122.24 ± 1.61	47.48 ± 1.92	69.20 ± 1.08

Note: The values from the iColorpalette website are also listed as reference.

close alignment with the $y = x$ line indicates the agreement between these two methods. Furthermore, readings by the Color Grab app on the Android phone are

consistent with the references, showing deviations of less than five for four colors, namely, Yellow ($H = 50$), Blue ($H = 200$), Pink ($H = 338$), and Red ($H = 357$). As shown in the scatter plots in Figure 5(b), significant deviations from the $y = x$ line occur only in the measurements for Gray and Green.

For S values, both the iPhone (using the Colorimeter app) and the Android phone (using the Color Grab app) accurately match the S value of the Blue card reference (100%). However, the PineTools program extracts a lower S value from this card. Deviations from the reference values are evident in other color cards, with all three measurement methods tending to overestimate S values for highly saturated colors, such as the Red, Yellow, and Orange cards. For Gray and Purple cards, the iPhone still overestimates even these lower S values, while readings from the Android phone align more closely with reference values.

The variations in V values are smaller than those observed for S values. For the Black and Green cards, a two-sample t-test revealed no significant difference between the readings obtained from the Colorimeter app and the PineTools program (p -value > 0.05). However, measurements from images taken with the iPhone exhibit lower accuracy than those obtained using the Color Grab app on the Android phone. The readings by the Color Grab app are consistent with the reference values, with only the Black card showing a large discrepancy. The

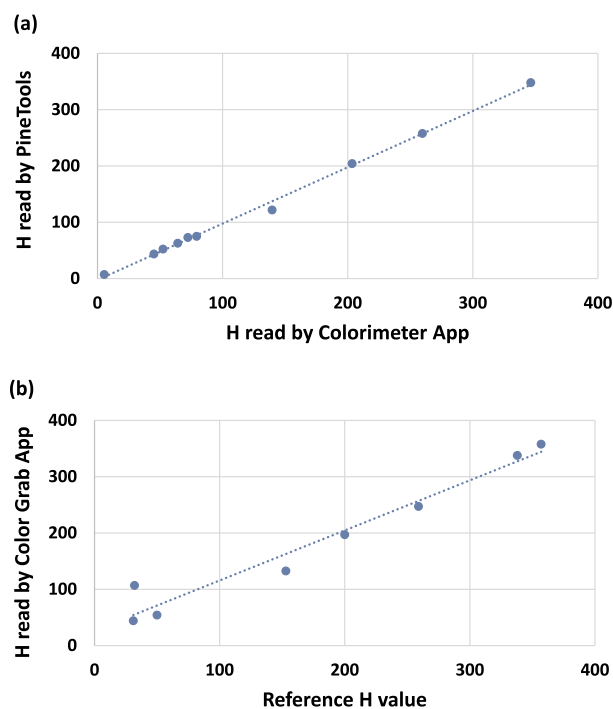
**Figure 5.** Plot of H values measured by (a) the PineTools program on the computer against the Colorimeter app on the iPhone and (b) the Color Grab app on the Android phone against reference values from the iColorpalette website.

Table 3. CIELAB parameters measured by the Color Grab app on the Android phone and the portable colorimeter, compared to reference values from the iColorpalette website.

Card color	L*a*b*	Portable Colorimeter		Color grab app	
	Reference values	L*a*b* values	ΔE_{ab}^*	L*a*b* Values	ΔE_{ab}^*
Black: Pantone 6 U	33.26, 1.03, -4.98	1.87, -0.39, -1.41	47.3419	40.29, -0.48, -1.80	36.5129
White: Plain Paper	—	89.83, -8.13, 7.19	—	99.99, -0.71, 1.98	—
Gray: Pantone 400 U	78.36, 0.86, 4.25	71.32, -6.52, 6.17	10.3791	89.08, 1.42, 4.25	10.7378
Red: Pantone 032 U	58.07, 63.48, 31.79	37.84, 58.90, 21.65	23.0864	64.22, 67.89, 11.53	21.6322
Green: Pantone 354 U	60.91, -54.76, 29.54	51.34, -41.03, 26.83	16.9519	74.49, -39.91, 24.51	20.7376
Blue: Pantone Process Blue	52.99, -4.78, -44.04	44.12, -10.57, -32.40	15.7370	67.92, -28.95, -33.25	30.3880
Yellow: Pantone 108 U	88.56, -4.06, 79.49	80.30, -12.99, 49.07	32.7598	94.93, 25.77, 73.37	31.1119
Pink: Pantone 212 U	64.01, 56.84, -0.45	55.29, 40.38, 1.17	18.6956	73.83, 57.39, -19.32	21.2815
Orange: Pantone 137 U	73.76, 28.05, 65.45	69.61, 3.46, 42.07	34.1817	85.96, 38.53, 40.77	29.4573
Purple: Pantone 2655 U	61.91, 26.7, -37.94	46.35, 10.07, -24.29	26.5575	67.70, 12.73, -27.28	18.5017

White card shows a high V value (91.67%), with V values exceeding 63% recorded for all other color cards. In contrast to the trend observed for S values, smartphone colorimetry underestimates V values relative to the references, with the Colorimeter app reading of the Purple card being the only exception.

3.3. L*a*b* readings by a portable colorimeter and the Color Grab app on an Android phone

Table 3 presents the measured CIELAB parameters obtained using the portable colorimeter and the Color Grab app, alongside Pantone reference values sourced from the iColorpalette website. The Color Grab app on the Android phone demonstrates the ability to correctly differentiate L* values, with high values exceeding 90 for the White and Yellow cards and lower values below 70 for the Red, Blue, and Purple cards. The trends in the a* and b* parameters also align well with the references. Specifically, the a* values range from a minimum of -39.91 for the Green card to a maximum of 67.89 for the Red card. Regarding b* values, the Yellow card shows a significant value of 73.37, in stark contrast to the Blue card, which records a value of -33.25.

The associated ΔE_{ab}^* parameter, calculated relative to the Pantone references in Table 3, provides a single metric to evaluate the measurement accuracy. Lower ΔE_{ab}^* values correspond to higher accuracy, while higher values indicate greater deviation from the reference. Both measurement methods produce considerable ΔE_{ab}^* , with the largest discrepancy observed for the Black card. For eight other cards, the Color Grab app on the Android phone generates ΔE_{ab}^* values in the range of 11–31. The lowest ΔE_{ab}^* values are obtained from the Gray card, while significant deviations are found in vibrant or saturated colors, including Blue, Yellow, and Orange cards. These ΔE_{ab}^* values are comparable to those from the portable colorimeter. Interestingly, the Color Grab app outperforms the portable colorimeter in five of the color cards by producing lower ΔE_{ab}^* values,

demonstrating greater measurement accuracy in these cases.

3.4. Implications for archaeological research

In archaeological settings, smartphones offer several advantages, including portability, affordability, and the ability to capture high-resolution images. As demonstrated in this study, color values in RGB, HSV, and CIELAB color spaces can be extracted from captured images using mobile applications or computer software. This capability makes smartphone colorimetry a promising tool for on-site and non-invasive analysis of color changes in artifacts and materials under different environmental conditions, complementing traditional spectroscopic methods. Potential applications include monitoring the degradation of pigments, tracking fading in mural paintings or pottery, assessing restoration efforts, and characterizing the colors of ancient artifacts, to provide a rapid assessment in the fieldwork.

Despite its potential, smartphone colorimetry has notable limitations. The color values derived from smartphone images are influenced by several factors, including lighting conditions, camera spectral sensitivity, and internal image processing algorithms. Another significant limitation is the absence of standardization to ensure consistency across devices. Different smartphone models and applications can produce inconsistent results, complicating cross-platform comparisons and reducing the reliability of measurements.

To address these challenges, measurements should be conducted in a controlled lightbox to ensure uniform lighting conditions and a fixed distance between the camera and the object during image acquisition. Calibration with standard color cards can improve accuracy, but the device-dependence of RGB values remains a significant limitation. The initial phase of this study utilizes the Colorimeter app and the PineTools program to extract color values in the RGB and HSV color spaces from images of ten color cards taken by an iPhone. Despite the precision of this smartphone-based

colorimetry, the results highlight significant inaccuracies in color values across measurements. In the second phase, an Android phone generally offers improved accuracy in HSV color measurements. Unlike iPhones, many Android phones allow greater flexibility in controlling image acquisition parameters, such as exposure and white balance, which were manually set to ensure consistency during measurements.

Although the overestimated S values might limit the interchangeability of the devices, the H and V values extracted by an Android phone offer closer agreement to the reference values. The underestimating but consistency in V values is encouraging, as it is reliable for certain applications where brightness is critical. For instance, the V values could be used to monitor the degradation of pigments, characterize the lightness of pottery glazes, evaluate color fading in cultural heritage artifacts, and assess the brightness of painted surfaces on architectural remains or restoration efforts. The Android phone also provides comparable $L^*a^*b^*$ values to a portable colorimeter. This CIELAB color space is less device-dependent and has recently been employed in classifying the color of ceramics (Gebremariam et al. 2022; Sirisathitkul, Sirisathitkul, and Noonsuk 2024), highlighting its potential in a field-based colorimetry when properly calibrated.

4. Conclusion

This study explores standardized methodologies for measuring RGB, HSV, and CIELAB parameters using colorimetric apps on an iPhone and an Android phone. Both smartphone colorimetry platforms exhibit high precision, as evidenced by the small standard deviations of repeated measurements. However, deviations from Pantone reference values are observed across different parameters. RGB values obtained via the Colorimeter app on the iPhone align well with the colors of the ten standard cards but still show inaccuracies compared to the references. Analyzing imported images with the PineTools program on a computer produces similar RGB and H readings to those from the app, though minor color components are frequently overestimated. Across all three measurement methods, S values are consistently overestimated, especially for highly saturated colors. Compared to the iPhone, the Color Grab app on the Android phone demonstrates higher accuracy in extracting H and V values, with results more closely aligned with Pantone reference values. The reliability of V values suggests potential applications in archaeological research, where brightness plays a critical role. Additionally, the Android phone provides L^* , a^* , and b^* values comparable to a portable colorimeter,

emphasizing its viability as a practical tool for on-site colorimetry, provided proper calibration is performed.

Acknowledgments

ChatGPT (<https://chat.openai.com/>) was used in refining the manuscript writing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Yaowarat Sirisathitkul obtained her M.Sc. in computer science from the National Institute of Development Administration, Thailand, in 2004. She currently serves as an assistant professor in the School of Engineering and Technology at Walailak University, Thailand. Her research focuses on computer programming and simulation, digital image processing, machine vision, and machine learning. She has also been involved in integrating smartphone sensors into engineering education, with published works on smartphone-based digital image colorimetry in forensic science and cultural heritage studies.

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Chitnarong Sirisathitkul obtained his D.Phil. in 2000 from the University of Oxford, UK. He has been working at Walailak University, Thailand, since 2001, where he was awarded the Best Teacher in 2005 and the Best Researcher in 2017. He is currently an associate professor and head of the Division of Physics at the School of Science. His publications in Scopus-indexed journals cover topics such as magnetic materials, nanoparticles, and traditional ceramics.

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